Scale-Bridging Techno-Economic, Microclimatic, and Open-Quantum Assessment of Semi-Transparent Organic Agrivoltaics for Sub-Saharan Perennial Crops

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Abstract

Semi-transparent organic photovoltaics (ST-OPVs) are promoted for tropical agrivoltaics assuming spectral filtering cools crops without silicon's light penalty and modulates photosynthetic quantum coherence. We test both premises via a matched-shade counterfactual model across Cameroon's Arabica coffee, Penja white pepper, and highland tea chains, coupling PM6:Y6 transfer-matrix optics, 8760 h ERA5–FvCB–FAO-56 physiology, battery dispatch, replacement-aware LCA, and $N = 10\,000$ Saltelli–Sobol draws alongside steady-state bounds on three 14-site LHCII Hamiltonians. Shapley attribution exposes a seasonal inversion: dry-season cooling lifts yields (8.8 to 30.2 %), whereas monsoon light deficits turn annual yields negative in coffee (−7.08 %) and tea (−3.35 %) at 40.6 % shade, leaving rainfed pepper positive (+7.43 to +8.16 %). At matched shade, ST-OPV and c-Si crop yields differ by ≤ 1.1 percentage points and spectral selectivity adds −0.2 to −0.7 %: shade fraction governs response. Tea storage lifts solar fraction to 99.9 %. Dual Sobol analysis separates bankability (tariffs and WACC govern $\mathrm{NPV_{OPV}}$) from technology choice: baseline ST-OPV (35 USD m^{-2}, $T_{80} = 5$ yr) trails c-Si NPV in 77 to 89 % of draws (75 to 88 % at zero carbon). Carbon parity needs $T_{80} \geq 7.4$ to 8.1 yr, while economic parity (pepper only) needs 15 USD m^{-2} modules with $T_{80} \geq 16.3$ yr. Quantum shifts stay $\leq 3.4 \times 10^{-5}$ ($\leq 2.1 \times 10^{-3}$ HEOM envelope, $S_{Ti} \leq 2.6 \times 10^{-7}$). Module cost, stability, and shade timing govern deployment.

Keywords: PM6:Y6 transfer-matrix optics, Cameroon highlands, Penja white pepper, Seasonal regime inversion, Environmental carbon payback, Sobol global sensitivity

1. Introduction

Dual land-use intensification via agrivoltaic (AV) co-production offers a concrete pathway to pair rural clean electrification and agricultural modernization with climate adaptation for heat-stressed perennial crops [1, 2, 3, 4, 5, 6, 7, 8]. Across Sub-Saharan Africa and humid–dry tropical highlands worldwide, high-value perennial cash crops—such as highland Arabica coffee, specialty spices, and orthodox tea—sustain smallholder cooperatives exposed both to intensifying dry-season vapor pressure deficits (VPD) and to unreliable, diesel-backed post-harvest processing grids. Unlike temperate annuals or arid-zone row crops, however, tropical highland perennials face a seasonal optical dilemma: extreme dry-season radiative heat (2.8 to 4.9 kPa VPD) invites canopy shading, whereas persistent wet-season monsoon cloudiness (0.42 to 0.50 diffuse sky fraction) risks light starvation under year-round structures. Conventional elevated

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opaque crystalline silicon (c-Si) arrays deliver low levelized cost of electricity (LCOE) [9, 10, 11], yet require heavy galvanized steel substructures for farm clearance ($\geq 3.5\,\mathrm{m}$) and cast spectrally neutral shadows that alter soil moisture, microclimate telemetry, and monsoon-season photosynthesis [12, 13, 14, 15, 16, 17, 18].

Semi-transparent organic photovoltaics (ST-OPVs) based on narrow-bandgap non-fullerene acceptors (NFAs) such as Y6 blended with wide-bandgap polymer donors such as PM6 [19, 20, 21, 22, 23, 24] have been advocated as spectrally tailored AV canopies [25, 26]. By absorbing ultraviolet and near-infrared photons (700 to 950 nm) while transmitting photosynthetically active radiation (PAR, 400 to 700 nm) on lightweight tensile structures, ST-OPVs in principle mitigate PAR competition while lowering structural balance-of-system (BOS) steel requirements [27, 28]. Simultaneously, comparison between photosynthetic and photovoltaic conversion efficiencies [29] and ultrafast spectroscopic evidence of transient coherence in photosynthetic light-harvesting complex II (LHCII) [30, 31, 32, 33, 34, 35, 36] have prompted speculation on whether spectrally filtered canopies alter primary excitonic energy transfer (EET) efficiency [37, 38].

A critical synthesis of existing literature (Table 1) reveals three cross-scale methodological gaps that prevent rigorous engineering and policy decisions for tropical ST-OPVs. First, seasonal regime inversion has rarely been quantified against matched-shade opaque counterfactuals. Most process-based agrivoltaic models focus on temperate or dry summer annuals [1, 9, 13, 14], whereas greenhouse ST-OPV demonstrations [25, 26] emphasize controlled transmittance without hourly tropical water–energy coupling. Resolving whether fixed shading yields a net annual benefit or penalty across humid–dry perennials—and isolating spectral selectivity from total shade fraction—requires hourly coupling of leaf energy balance, Farquhar–von Caemmerer–Berry (FvCB) photosynthetic kinetics, photoinhibition relief, and FAO-56 root-zone water conservation evaluated directly against both traditional agroforestry (S_1) and elevated c-Si at equal shade [39, 40, 13, 41, 14].

Table 1: Methodological positioning of the present scale-bridging assessment relative to established agrivoltaic, semi-transparent OPV, techno-economic, and open-quantum literature streams.

Literature stream	**Climate / crop scope**	**Optical canopy model**	**Dispatch, TEA & LCA**	**Quantum dynamics**	**Primary scope & gap**
Opaque elevated AV modeling [1, 9, 13, 14, 17]	Temperate / arid annuals & perennials	Opaque neutral shade projection	Agronomic yield & LER; limited BESS/LCA	None	Establishes crop water–light mechanics; omits NFA spectral optics
ST-OPV greenhouses & crops [25, 26, 24]	Controlled greenhouses / lettuce, crops	Wavelength-selective TMM (PM6:Y6)	Thermal energy balance; zero cooperative TEA/LCA	None	Demonstrates spectral selectivity; lacks humid–dry outdoor seasonal testing
AV regional TEA, LCA & MC [28, 10, 11]	Temperate US / Europe / utility grids	Fixed or tilted opaque/bifacial c-Si	Full TEA/LCA or European MC; no tropical dispatch	None	Identifies CAPEX/BOS dominance; excludes tropical OPV replacement
Open-quantum LHCII EET [42, 37, 38, 36]	Molecular pigment complexes	Idealized or unfiltered pumping	None	Pulsed HEOM / MesoHOPS vs. CW NESS	Defines quantum–classical physics; uncoupled from farm economics
Present study	*Tropical wet–dry perennials (3 crops)*	*Calibrated PM6:Y6 TMM vs. matched c-Si*	*Hourly BESS dispatch, dual-FU LCA, $N = 10\,000$ Sobol*	*3 LHCII Hamiltonians, HEOM/MesoHOPS + NPV*	*Falsifiable counterfactual test across 5 scales*

Second, module techno-economics, replacement-aware life-cycle assessment (LCA), and cooperative post-harvest dispatch are seldom integrated within a single decision framework [28, 11]. Because current organic photovoltaic lifetimes remain finite ($T_{80} = 5$ to $10\,\mathrm{yr}$) [27, 43, 44, 28], periodic module replacement elevates embodied carbon per functional kilowatt-hour while eroding cumulative cash flow. These penalties must be weighed jointly against site-specific processing demand profiles—such as daytime spice drying versus continuous 24 h tea

withering requiring battery energy storage (BESS)—under the zero-subsidy, zero-carbon-price baseline typical of rural agricultural cooperatives, and tested separately for both standalone bankability ($\mathrm{NPV_{OPV}}$) and technology selection ($\Delta\mathrm{NPV} = \mathrm{NPV_{OPV}} - \mathrm{NPV_{Si}}$). Third, the quantum premise behind spectrally tailored canopies remains untested at agricultural system scale. Although 2D electronic spectroscopy and non-Markovian hierarchical equations of motion (HEOM/MesoHOPS) demonstrate fs–ps coherent dynamics under pulsed excitation [45, 46, 47, 48, 49, 50, 36], prompting hypotheses that spectral filtering could alter excitonic energy transfer [37, 38], natural continuous-wave (CW) sunlight drives incoherent absorption into a non-equilibrium steady state (NESS) [42, 37]. Propagating a multi-Hamiltonian CW steady-state bound through crop assimilation into farm net present value (NPV) tests whether ST-OPV canopy design requires quantum-efficiency objectives or can rely exclusively on classical radiometric metrics.

To resolve these coupled physical and economic trade-offs, this study addresses two central questions across western Cameroon's three flagship perennial chains—Arabica coffee (*Coffea arabica*, Foumbot), Protected Geographical Indication (PGI) Penja white pepper (*Piper nigrum* L., Njombe-Penja), and highland tea (*Camellia sinensis*, Djuttitsa)—whose VPD, diffuse cloudiness, and processing load regimes span archetypal wet–dry tropical highlands: (i) what dynamic balance of leaf cooling, VPD conservation, monsoon PAR loss, and bounded spectral or quantum redistribution governs rainfed crop yields; and (ii) under hourly tropical seasonality, cooperative processing loads, and zero-subsidy baseline economics, how do PM6:Y6 ST-OPV and elevated opaque c-Si compare in crop yield relative to open sun, land equivalent ratio (LER), 20-year array NPV per installed hectare, and life-cycle carbon intensity, and how do these outcomes vary with crop, dispatch, module cost, and stability? Furthermore, when bounded across three independent LHCII Hamiltonians under incoherent solar illumination, does filtered excitonic coherence contribute materially to modeled agronomic yield or financial return? By separating shade from spectral effects through Shapley attribution, coupling replacement-aware LCA with Dual Sobol parity frontiers, and propagating a deliberately conservative quantum lever through the same modeling chain, this work evaluates spectral enhancement and quantum modulation as model-based, testable system-scale hypotheses.

Nomenclature

Symbol / Acronym	Definition (Units)	Symbol / Acronym	Definition (Units)
APT	Average photosynthetic transmittance (400 to 700 nm, %)	S_i, S_{Ti}	First-order and total-order Sobol indices (–)
AWC	Root-zone available water capacity (mm)	T_{80}	Time to 80 % initial PV PCE (yr)
BESS	Battery energy storage system (LFP, kWh)	$T_{\mathrm{leaf}}, T_{\mathrm{PV}}$	Leaf and PV module temperatures (°C)
BOS	Balance-of-system structural & electrical cost ($\mathrm{USD\,m^{-2}}$)	$T_{\mathrm{OPV}}^{\mathrm{beam}}, T_{\mathrm{OPV}}^{\mathrm{diff}}$	Normal beam and diffuse transmittance (–)
$C_{\mathrm{mod}}^{\mathrm{OPV}}, C_{\mathrm{BOS}}^{\mathrm{c\text{-}Si}}$	OPV module and c-Si BOS unit costs ($\mathrm{USD\,m^{-2}}$)	$V_{c\max 25}, J_{\max 25}$	Max carboxylation and electron transport rates ($\mu\mathrm{mol\,m^{-2}\,s^{-1}}$)
CW / NESS	Continuous-wave / non-equilibrium steady state	$\mathrm{VPD_{95}}$	95th-percentile dry-season vapor pressure deficit (kPa)
ePAR	Extended PAR (400 to 750 nm)	WUE	Crop water-use efficiency ($\mathrm{kg\,m^{-3}}$)
$F_{\mathrm{lp}}, F_{\mathrm{sky}}$	Low-PPFD hour fraction and sky view factor (–)	$\Delta\eta_{\mathrm{NESS}}$	Coherent vs. Pauli EET efficiency shift (–)
GCR	Array ground coverage ratio (–)	$\Delta T_{\mathrm{leaf}}, \Delta Y$	Leaf cooling (°C) and crop yield anomaly (%)
$I_{\mathrm{PI}}, I_{\mathrm{sat}}$	Photoinhibition and saturation PPFD ($\mu\mathrm{mol\,m^{-2}\,s^{-1}}$)	λ_D, γ_D	Drude–Lorentz bath reorganization and cutoff ($\mathrm{cm^{-1}}$)
$k_{\mathrm{RC}}, k_{\mathrm{NPQ}}$	Reaction-center trapping ($\mathrm{ps^{-1}}$) and NPQ rate ($\mathrm{ns^{-1}}$)	$\phi_{\mathrm{PAR/PI}}, \phi_{\mathrm{therm}}$	Shapley PAR and thermal yield contributions (%)
LER, LSF, LUE	Land equivalent ratio, load solar fraction, light util. eff.	$\phi_{\mathrm{VPD/H_2O}}, \phi_{\mathrm{spec+Q}}$	Shapley water and spectral+quantum contributions (%)
NPV, IRR, LCOE	Net present value ($\mathrm{USD\,ha^{-1}}$), internal rate of return (%), LCOE ($\mathrm{USD\,kWh^{-1}}$)	R:FR	Red-to-far-red photon flux ratio (–)

2. Materials, site calibration, and scale-bridging methodology

All physical, botanical, optical, and economic inputs are consolidated from peer-reviewed physiological datasets and primary regional surveys summarized in Supporting Material S3.

2.1. Counterfactual design and analytical boundaries

Because this investigation evaluates a counterfactual engineering problem without paired field agrivoltaic measurements at the three Cameroonian sites, all crop responses represent conditional simulations generated by literature-calibrated process models driven by the full-year 2023 ERA5 reanalysis. Accordingly, conclusions are framed conditionally on model structure

and parameter bounds across six counterfactual configurations: an open-field unshaded control (S_0), a traditional shade-tree agroforestry reference (S_1), elevated opaque crystalline silicon counterfactuals at GCR = 40 % and 70 % (S_2(c-Si, GCR40) and S_2(c-Si, GCR70)), baseline PM6:Y6 ST-OPV canopies (T_{80} = 5 yr) spanning APT = 20 to 56 % and GCR = 50 to 85 % (denoted S_3(APT x, GCR y), with the balanced focal design S_3(APT44, GCR65) matching the $\sim$ 40 % shade of S_2(c-Si, GCR40) to isolate spectral selectivity from total shading), encapsulated long-lived ST-OPV variants (S_4, T_{80} = 15 to 25 yr), and dry-season supplementally irrigated ST-OPV arrays (S_5). Exact four-lever Shapley shares apportion modeled yield shifts among internal physical pathways rather than identifying econometric treatment effects from observational trial data. Throughout the global uncertainty propagation, outcomes are classified as robust when their sign persists across the Sobol parameter space and conditional when contingent on specific module-cost, operational-lifetime, or export-tariff thresholds. To evaluate the quantum biology premise conservatively, excitonic steady-state efficiency bounds are propagated one-to-one into crop assimilation across intervals up to 30 to 60 times wider than the computed shifts, so that negligible variance contribution provides strong evidence against macroscopic quantum leverage. Array NPV is expressed per hectare of agrivoltaic installation serving the cooperative processing load with surplus electricity exported to the grid unless explicitly operated off-grid.

2.2. Cameroon highland case studies and botanical distinctness

Table 2 summarizes the annual 2023 ERA5 meteorological baselines (8760 h UTC) [51] and calibrated FvCB–Medlyn ecophysiological parameters for the three Cameroon production basins.

Table 2: ERA5 2023 annual site climate summaries (8760 h per site) and calibrated FvCB–Medlyn ecophysiological parameters for the three Cameroon perennial systems.

Attribute	**Arabica Coffee** *Coffea arabica*	**Penja White Pepper** *Piper nigrum* L. (PGI)	**Highland Tea** *Camellia sinensis*
Location (ERA5 grid)	Foumbot (5.518° N, 10.597° E)	Njombe-Penja (4.675° N, 9.680° E)	Djuttitsa (5.448° N, 10.066° E)
Elevation	1044 m	273 m	1400 m
Annual T_{mean} $[T_{\mathrm{min}}, T_{\mathrm{max}}]$	22.21 °C [14.0, 35.0]	24.72 °C [19.8, 35.8]	19.86 °C [14.2, 31.0]
Annual GHI (kWh m^{-2} d^{-1} / yr)	5.382 (1964.6)	4.593 (1676.4)	4.900 (1788.6)
Diffuse fraction (Ann / Wet / Dry)	0.365/0.416/0.260	0.461/0.502/0.360	0.412/0.486/0.273
Daytime mean VPD (Dry p95 / max)	1.381 kPa (4.07/4.85)	0.890 kPa (2.83/4.75)	0.922 kPa (3.03/3.80)
Annual rainfall	1904.5 mm	2321.0 mm	2320.3 mm
Baseline clean yield (Y_0)	850 kg ha^{-1}	800 kg ha^{-1} (dry white)	1800 kg ha^{-1} (made tea)
Calibrated $V_{c\,\max 25}$ / $J_{\max 25}$ (µmol m^{-2} s^{-1})	78.5/142.6 (C_c sun) [52]	44.6/65.0 (C_i-basis) [53, 54]	48.0/88.0 [55, 56]
Photoinhibition PPFD threshold	1100 µmol m^{-2} s^{-1}	900 µmol m^{-2} s^{-1}	1200 µmol m^{-2} s^{-1}
Cooperative catchment	3000 ha (63.8 MWh yr^{-1})	2000 ha (72.0 MWh yr^{-1})	450 ha (307.8 MWh yr^{-1})
Processing demand (kWh t^{-1}, day share)	25.0 (85 %)	45.0 (85 %)	380.0 (55 %, 45 % night)

Foumbot coffee exhibits the sharpest dry-season radiative and evaporative stress (VPD_{95} = 4.07 kPa, annual GHI = 1964.6 kWh m^{-2} yr^{-1}), whereas Njombe-Penja experiences warmer ambient temperatures (24.72 °C mean) and persistent cloudiness (0.461 diffuse fraction). Djuttitsa tea combines cool elevation (1400 m) with high cooperative processing intensity (380 kWh t^{-1} electrical demand, 45 % overnight withering). For *Piper nigrum* in Njombe-Penja, we explicitly

enforce PGI Penja white pepper post-harvest yields ($\sim 24\,\%$ dry kernel recovery from red-ripe berries $= 800\,\mathrm{kg\,ha^{-1}}$, versus $\sim 34\,\%$ whole-dried black pepper), white farm-gate prices ($5000\,\mathrm{XAF\,kg^{-1}}$, PGI certified tier $14\,000\,\mathrm{XAF\,kg^{-1}}$), and retting/decortication/fan-drying electric load ($45.0\,\mathrm{kWh\,t^{-1}}$), plus avoided dead or living tutor staking ($350\,\mathrm{USD\,ha^{-1}\,yr^{-1}}$ overhead trellis substitution, bounds 200 to $550\,\mathrm{USD\,ha^{-1}\,yr^{-1}}$; Supporting Material Table S1).

2.3. Optical transfer-matrix optimization and canopy radiation

Semi-transparent PM6:Y6 stacks comprising glass/ITO (140 nm)/ZnO (30 nm)/PM6:Y6 (d_{act})/MoO_3 (10 nm)/Ag (d_{Ag})/MoO_3 (d_{cap}) are solved over $d_{\mathrm{act}} \in [40, 140]$ nm, $d_{\mathrm{Ag}} \in [8, 22]$ nm, and $d_{\mathrm{cap}} \in [0, 60]$ nm (120 grid stacks at 1 nm spacing) using 2×2 Abeles transfer matrices populated with variable-angle ellipsometry data and dispersive Johnson–Christy silver constants (Supporting Material Section S1) [19, 25, 24]. Unpolarized normal beam transmission $T_{\mathrm{OPV}}^{\mathrm{beam}}(\lambda, 0^\circ)$ and isotropic hemispherical diffuse transmission,

$$T_{\mathrm{OPV}}^{\mathrm{diff}}(\lambda) = 2 \int_0^{\pi/2} T_{\mathrm{OPV}}^{\mathrm{beam}}(\lambda, \theta) \sin\theta \cos\theta \, \mathrm{d}\theta, \tag{1}$$

are integrated hourly against ERA5 direct/diffuse irradiance splits while enforcing machine-precision optical photon conservation ($\max|R+T+A_{\mathrm{tot}}-1| = 4.44\times10^{-16}$). The coupled Pareto frontier maximizes Light Utilization Efficiency LUE = PCE × APT. Photomorphogenic canopy quality is tracked via extended-PAR (ePAR, 400 to 750 nm) transmittance relative to standard PAR and the red-to-far-red shift Δ(R:FR) [26, 57]. Opaque crystalline silicon counterfactuals S_2(c-Si, GCR40) and S_2(c-Si, GCR70) model 20.5 % STC efficiency with $-0.38\,\%\,^\circ\mathrm{C}^{-1}$ thermal derating (versus near-zero $-0.008\,\%\,^\circ\mathrm{C}^{-1}$ for PM6:Y6 NFAs) and spectrally neutral shadow projection.

2.4. Hourly microclimate, tri-crop physiology, and four-lever Shapley attribution

At each hour $t \in [1, 8760]$, sub-canopy incident direct and diffuse fluxes combine ground coverage ratio (GCR $\in [0.40, 0.85]$), row geometry view factor (F_{sky}), and TMM transmittance [14, 17]. Module temperature (T_{PV}) and sunlit/shaded leaf temperatures (T_{leaf}) satisfy coupled convective, longwave sky/module radiative, and latent Penman–Monteith energy balances [58, 12] (max energy residual $\leq 1.569 \times 10^{-5}\,\mathrm{W\,m^{-2}}$). Net leaf assimilation couples minimum Rubisco-limited (W_c) and RuBP-regeneration-limited (W_j) FvCB rates [39], calibrated temperature acclimation, high-PPFD photoinhibition above crop-specific thresholds (Table 2), and Medlyn stomatal conductance [40] coupled to daily FAO-56 root-zone soil water balance (AWC = 160 mm rainfed vs. S_5 supplemental irrigation) [13]. Including the traditional shade-tree agroforestry baseline (S_1, at crop-calibrated diffuse/direct shade fractions of 32.5 % in coffee, 27.5 % in pepper, and 40.0 % in tea with zero PV self-heating) allows engineered PV canopies to be benchmarked against living smallholder practice alongside open sun S_0 and matched-shade c-Si.

To separate physical mechanisms relative to the open-field baseline S_0, annual yield shifts ΔY are decomposed across four model levers—(1) PAR attenuation & photoinhibition relief ($\phi_{\mathrm{PAR/PI}}$), (2) leaf thermal cooling ΔT_{leaf} (ϕ_{therm}), (3) stomatal VPD & soil-water conservation ($\phi_{\mathrm{VPD/H_2O}}$), and (4) spectral ePAR/R:FR quality & quantum NESS EET perturbation ($\phi_{\mathrm{spec+Q}}$)—by evaluating all $2^4 = 16$ coalition combinations. Because the levers interact non-additively, Shapley averaging is used to share the interaction terms fairly:

$$\phi_i = \sum_{S \subseteq \mathcal{N}\setminus\{i\}} \frac{|S|!\,(4-|S|-1)!}{4!} \Big[Y(S \cup \{i\}) - Y(S) \Big], \qquad \sum_{i=1}^{4} \phi_i = \Delta Y_{\mathrm{annual}}, \tag{2}$$

guaranteeing machine-precision closure ($\max|\sum \phi_i - \Delta Y| \leq 7.105 \times 10^{-15}\,\%$).

2.5. Microscopic open-quantum LHCII model and scale propagation

To quantify whether sub-canopy spectral filtering alters primary exciton transport inside crop photosystems, we solve Bloch–Redfield partial-secular versus diagonal Pauli non-equilibrium steady states ($\mathcal{L}\rho_{\rm ss} = 0$) [59, 60, 31, 61, 62, 63, 38] across three independent 14-pigment LHCII Hamiltonians (H1 Novoderezhkin 2005/2011, H2 Müh 2010, H3 MMSIC-MD 2025 [64], consistent with recent spinach PSII-LHCII exciton modeling [65]) with PDB 1RWT/PDB 2BHW non-orthogonal dipoles, Renger spectral density ($\lambda_D = 50.87\,\mathrm{cm}^{-1}$, $\gamma_D = 50\,\mathrm{cm}^{-1}$), $T = 295\,\mathrm{K}$, $N = 200$ disorder draws, open reaction-center trapping $k_{\rm RC} \in [0.0033, 0.5]\,\mathrm{ps}^{-1}$, and NPQ quenching $k_{\rm NPQ} \in \{0, 1.5\}\,\mathrm{ns}^{-1}$. Direct non-Markovian checks use exact QuTiP-BoFiN HEOM on 4-site sub-clusters [45, 46, 49] and full 14-site MesoHOPS 1.8.0 pulsed propagation [47, 48, 50] ($N = 48$ seeds, 2000 fs; max trace-norm error $\leq 1.11 \times 10^{-15}$; Supporting Material Section S2).

2.6. Cooperative dispatch, dual-FU LCA, techno-economics, and Sobol propagation

Hourly PV generation serves cooperative processing loads (Table 2) across three battery dispatch configurations: daytime-only operation without storage, a 2 h lithium-iron-phosphate (LFP) buffer, and an overnight-withering LFP system (LFP CAPEX 220 USD kWh^{-1}, Douala commercial quotes 195 to 280 USD kWh^{-1}; max hourly dispatch energy-balance residual $\leq 4.42 \times 10^{-9}$ kWh). The financial functional unit is one hectare of installed agrivoltaic array. Factory demand is derived from crop production over the cooperative catchment in Table 2 and allocated to the modeled array; generation beyond that allocated demand is assumed exportable at the stated grid tariff. This is a modeled dispatch and accounting boundary, not evidence of a built interconnection or measured cooperative electricity sharing. Financial evaluation spans a 20 yr horizon (WACC = 10 % baseline, 6 to 15 % stress) strictly at 0 USD $\mathrm{tCO_2e}^{-1}$ carbon price and 0 % CAPEX grant, reporting NPV, internal rate of return (IRR), levelized cost of electricity (LCOE), and discounted payback period (DPBP). Baseline ST-OPV economics assume 35 USD m^{-2} module CAPEX (10 to 45 USD m^{-2} benchmark bounds covering R2R 9.2 to 15 USD m^{-2} [27, 43]), 20 USD m^{-2} lightweight tensile BOS, and $T_{80} = 5$ yr module replacement, benchmarked against elevated c-Si (45 USD m^{-2} module, 55 USD m^{-2} steel BOS, 25 yr life). Life-cycle emissions follow ISO 14040/14044 with dual functional units ($\mathrm{kgCO_2e\,ha^{-1}\,yr^{-1}}$ annualized over 20 yr and $\mathrm{gCO_2e\,kWh^{-1}}$, OPV embodied module carbon 20.0 $\mathrm{kgCO_2e\,m^{-2}}$ vs. c-Si 105.0 $\mathrm{kgCO_2e\,m^{-2}}$, grid displacement factor Cameroon hydro/thermal mix). Global uncertainty analysis uses Saltelli/Jansen quasi-Monte Carlo estimators [66] with $N = 10\,000$ base samples over 12 uncertain inputs ($N(12+2) = 140\,000$ model evaluations per crop): OPV module cost (10 to 45 USD m^{-2}), T_{80} (3 to 25 yr), OPV and c-Si BOS, WACC, diesel and export tariffs, LFP cost, crop price and yield multipliers, carbon price (0 to 25 USD $\mathrm{tCO_2e}^{-1}$), and the quantum lever $\Delta\eta_{\rm NESS} \in [-0.02, +0.10]$ %, which is about 30 times wider than the computed NESS bound and enters crop revenue one-to-one. To disentangle cooperative bankability from technology selection, Sobol variance indices are evaluated jointly for standalone ST-OPV returns ($\mathrm{NPV_{OPV}}$) and comparative technology parity ($\Delta\mathrm{NPV} = \mathrm{NPV_{OPV}} - \mathrm{NPV_{Si}}$); robustness reruns pin the carbon price at zero and widen the quantum lever to ± 0.21 % (the exact-HEOM envelope, Section 4).

3. Results

3.1. Optical Pareto frontier and photomorphogenic spectrum shaping

Sweeping active, metal, and capping thicknesses establishes a sharp PCE–APT Pareto front bounded between APT = 14.54 % and 52.06 % ($\mathrm{LUE_{max}} = 4.424$ % across the Pareto-optimal subset; 4.486 % grid peak; Figure 1a), while preserving photon conservation to machine precision ($\max|R + T + A_{\rm tot} - 1| = 4.44 \times 10^{-16}$) and reproducing the experimental PM6:Y6 benchmark [24] on an opaque reference cell ($d_{\rm act} = 75$ nm, $d_{\rm Ag} = 100$ nm: $J_{\rm sc} = 25.70\,\mathrm{mA\,cm^{-2}}$,

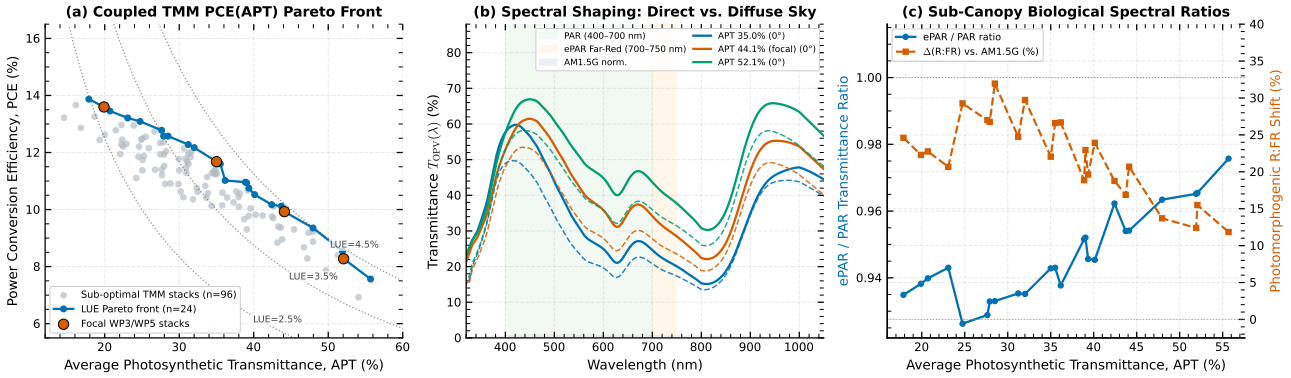


Figure 1: Transfer-matrix optical design of PM6:Y6 semi-transparent organic photovoltaic (ST-OPV) canopies. (a) Coupled PCE(APT) Pareto frontier across 120 simulated stacks ($n = 24$ Pareto optimal, $\mathrm{LUE}_{\max} = 4.424\,\%$ in library), highlighting the four focal agrivoltaic designs (APT $\approx$ 20 %, 35 %, 44.1 %, 52.1 %). (b) Direct normal (0°, solid) versus diffuse hemispherical (dashed) spectral transmittance across PAR (green shading) and far-red ePAR bands (orange shading). (c) Sub-canopy ePAR/PAR transmittance ratio and photomorphogenic Δ(R:FR) shift relative to AM1.5G.

PCE = 15.50 % vs. measured 26.36 mA cm^{-2} and 16.27 %). Selecting the focal balance stack APT44 ($d_{\mathrm{act}} = 60\,\mathrm{nm}$, $d_{\mathrm{Ag}} = 12\,\mathrm{nm}$, $d_{\mathrm{cap}} = 35\,\mathrm{nm}$) achieves APT = 44.12 % and PCE = 9.93 % at AM1.5G, whereas thinner silver/active stacks raise transparency to APT = 52.1 % (PCE $\approx$ 8.3 %). Spectral transmittance curves (Figure 1b) show pronounced blue–green transmission (420 to 540 nm, peaking above 61 % for APT44) paired with strong PM6:Y6 absorption across the red–NIR region (600 to 850 nm). Crucially, hemispherical diffuse transmittance sits 5 to 10 percentage points below 0° normal beam transmittance due to higher oblique Fresnel reflection—a non-negligible penalty during humid Cameroon monsoons where diffuse sky fractions reach 41.6 to 50.2 %. Across Pareto designs, differential attenuation near Y6's 700 nm band edge keeps ePAR/PAR $\in [0.926, 0.976]$ while moderately shifting transmitted R:FR by +11 to +32 % (Figure 1c).

3.2. Microclimatic cooling, seasonal regime inversion, and four-lever attribution

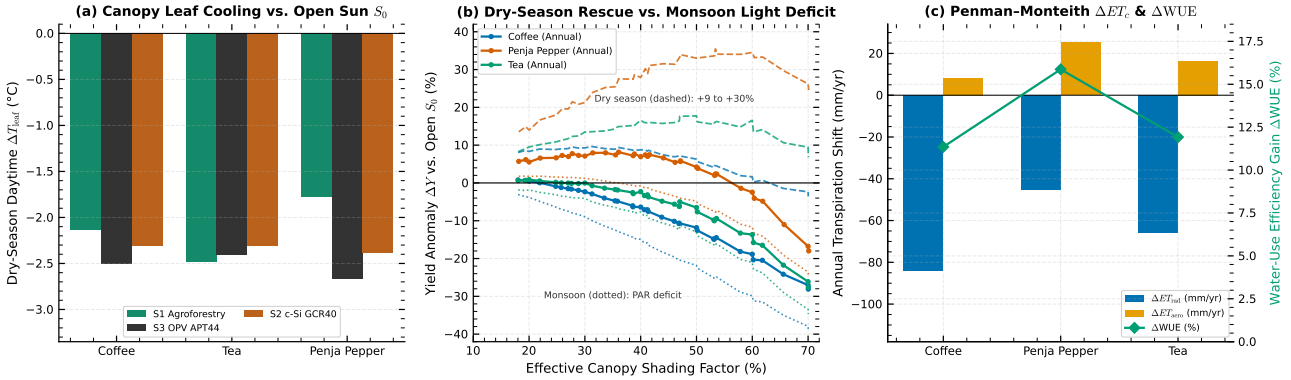


Figure 2: Hourly ERA5 microclimate and seasonal agronomic divergence across Cameroon highlands. (a) Dry-season daytime leaf cooling (ΔT_{leaf}) relative to open sun S_0 across agroforestry (S_1), c-Si (S_2), and ST-OPV (S_3). (b) Annual (solid), dry-season (dashed), and monsoon wet-season (dotted) crop yield anomalies versus effective canopy shading factor. (c) Penman–Monteith radiative (ΔET_{rad}) versus aerodynamic ($\Delta ET_{\mathrm{aero}}$) transpiration shifts and water-use efficiency gains (ΔWUE) under the focal balanced design S_3(APT44, GCR65).

Figure 2 and Table 3 demonstrate a decisive seasonal regime inversion across all three perennial crops. During intense dry-season midday heat, partial shading cuts canopy radiative load and cools leaves by −1.51 to −2.67 °C rainfed (ΔT_{leaf}) and by −3.10 to −3.45 °C with supplemental irrigation (S_5; Figure 2a). Simultaneously, spectrally selective transmittance lowers daytime

module self-heating on S_3(APT44, GCR65) to $\Delta T_{\mathrm{PV}} = 3.56$ to $4.29\,^{\circ}\mathrm{C}$ above ambient versus 7.73 to $9.47\,^{\circ}\mathrm{C}$ on opaque S_2(c-Si, GCR40), curbing NFA efficiency derating relative to c-Si even though both systems produce similar leaf cooling (-2.40 to $-2.67\,^{\circ}\mathrm{C}$ vs. -2.31 to $-2.38\,^{\circ}\mathrm{C}$). Because Foumbot ($\mathrm{VPD}_{95} = 4.07\,\mathrm{kPa}$) and Njombe-Penja ($T_{\max} = 35.8\,^{\circ}\mathrm{C}$) frequently cross stomatal closure and photoinhibition thresholds ($I_{\mathrm{PI}} = 900$ to $1100\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$), shading delivers strong dry-season yield gains under S_3(APT44, GCR65) of $+8.82\,\%$ (coffee), $+15.98\,\%$ (tea), and $+30.24\,\%$ (Penja white pepper; Figure 2b, dashed curves). Simultaneously, Penman–Monteith transpiration declines (-7.28 to $-16.56\,\%$) because radiative suppression ($\Delta ET_{\mathrm{rad}} = -83.98\,\mathrm{mm\,yr^{-1}}$ in coffee, $-65.72\,\mathrm{mm\,yr^{-1}}$ in tea, and $-45.29\,\mathrm{mm\,yr^{-1}}$ in pepper) dominates modest rainfed aerodynamic increases ($\Delta ET_{\mathrm{aero}} = +8.22$, $+16.41$ and $+25.18\,\mathrm{mm\,yr^{-1}}$, respectively), lifting crop water-use efficiency (ΔWUE) by $+11.35\,\%$ in coffee, $+11.92\,\%$ in tea, and $+15.87\,\%$ in Penja white pepper (Figure 2c); adding dry-season irrigation in S_5 restores stomatal conductance and drives $\Delta ET_{\mathrm{aero}}$ up to $+43.26$ to $+63.17\,\mathrm{mm\,yr^{-1}}$.

However, overcast monsoon conditions invert this mechanism: 36.1 to $41.6\,\%$ of daytime hours under GCR $= 65\,\%$ fall into light-limited PAR regimes (F_{lp}), inflicting wet-season yield losses of $-15.88\,\%$ in Arabica coffee and $-8.84\,\%$ in highland tea (Figure 2b, dotted curves). Benchmarked against traditional agroforestry (S_1, which achieves annual shifts of $-2.35\,\%$ in coffee at $32.5\,\%$ shade, $+7.96\,\%$ in pepper at $27.5\,\%$ shade, and $-1.81\,\%$ in tea at $40.0\,\%$ shade), rainfed coffee under S_3(APT44, GCR65) drops $-7.08\,\%$ annually and tea drops $-3.35\,\%$ (near neutral at $-1.21\,\%$ and $+0.01\,\%$ only at sparse S_3(APT56, GCR50)), whereas PGI Penja white pepper sustains net annual yield gains of $+7.27\,\%$ under S_3(APT56, GCR50), $+8.16\,\%$ under S_3(APT52, GCR65), and $+7.43\,\%$ under S_3(APT44, GCR65) (Table 3). Heavy shading ($70\,\%$ effective shade, S_2(c-Si, GCR70) or S_3(APT20, GCR85)) imposes severe -16.8 to $-28.1\,\%$ yield penalties across every species. Adding dry-season supplemental irrigation in S_5(APT44, GCR65, irrig.) rescues coffee to $+0.22\,\%$ ($+38.23\,\%$ dry boost), elevates tea to $+2.47\,\%$ ($+41.38\,\%$ dry), and propels Penja white pepper to $+12.25\,\%$ ($+49.68\,\%$ dry).

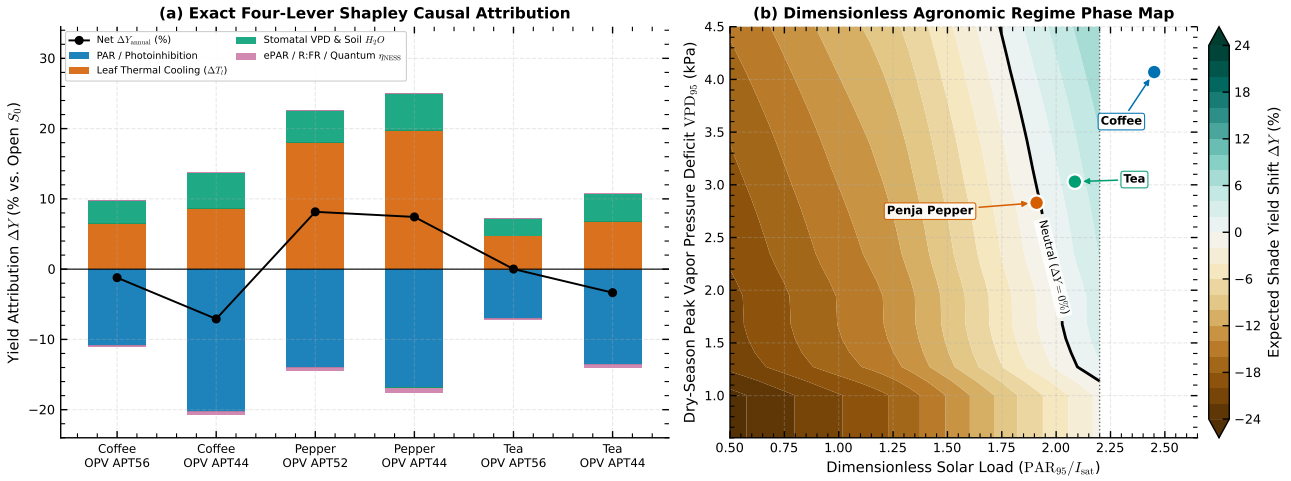


Figure 3: Causal mechanism and climate-envelope mapping. (a) Exact 16-coalition Shapley attribution partitioning total annual yield shift ΔY (black line) into PAR/photoinhibition, thermal leaf cooling, VPD/soil-water conservation, and spectral+quantum levers (residual $\leq 7.1 \times 10^{-15}\,\%$). (b) Dimensionless agronomic regime phase diagram spanned by normalized dry-season solar load ($\mathrm{PAR}_{95}/I_{\mathrm{sat}}$) and VPD_{95}, locating all three Cameroon crops near/above the neutral shade-benefit threshold.

Exact four-lever Shapley attribution (Figure 3a) apportions these modeled yield responses across physical mechanisms. For Penja white pepper under S_3(APT52, GCR65) ($+8.16\,\%$), thermal leaf cooling ($+17.99\,\%$) and stomatal/soil moisture conservation ($+4.64\,\%$) outweigh the PAR reduction penalty ($-13.93\,\%$), while optical spectral quality plus quantum NESS ($\phi_{\mathrm{spec+Q}} = -0.55\,\%$) play a minor sub-percent role. Conversely, in Foumbot coffee under

S_3(APT44, GCR65) (−7.08 %), the PAR penalty (−20.31 %) dominates combined thermal (+8.58 %) and soil-water (+5.11 %) gains. Directly comparing ST-OPV and elevated c-Si at matched ∼ 40 % shade (S_3(APT44, GCR65) vs. S_2(c-Si, GCR40); Table 3) reveals nearly indistinguishable annual crop yields—differing by at most 1.05 percentage points (−7.08 % vs. −6.38 % in coffee, +7.43 % vs. +6.95 % in pepper, −3.35 % vs. −2.30 % in tea). Across all ST-OPV designs (APT56–APT44), $\phi_{\text{spec+Q}}$ contributes only −0.21 to −0.67 % (slightly negative because PM6:Y6 band-edge absorption reduces the transmitted ePAR/PAR ratio below open sunlight), indicating within the coupled model that total shade fraction and wet–dry seasonality, rather than spectral selectivity, govern agronomic performance. Mapping peak irradiance normalized by species saturation (I_{sat}) against VPD_{95} (Figure 3b) highlights the narrow envelope governing annual rainfed viability without dynamic tracking.

Table 3: Focal seasonal agronomic performance, exact four-lever Shapley decomposition (ϕ_i), water-use shift (ΔWUE), and annual PV yield across Cameroon perennials.

Crop	**Scenario**	**Shade** (%)	ΔY_{ann} (%)	ΔY_{dry} (%)	ΔY_{wet} (%)	$\phi_{\text{PAR/PI}}$ (%)	ϕ_{therm} (%)	$\phi_{\text{H}_2\text{O}}$ (%)	ΔWUE (%)	**PV** (MWh ha^{-1})
Coffee	S3 (APT56, GCR50)	25.7	−1.21	+9.02	−6.96	−10.82	+6.45	+3.37	+7.80	717.0
Coffee	S3 (APT44, GCR65)	40.6	−7.08	+8.82	−15.88	−20.31	+8.58	+5.11	+11.35	1223.3
Coffee	S5 (APT44, GCR65, irrig.)	40.6	+0.22	+38.23	−15.88	−21.38	+10.81	+11.25	+0.15	1223.3
Coffee	S2 (c-Si, GCR40)	40.0	−6.38	+8.73	−14.92	−19.24	+8.14	+4.71	+9.56	1479.0
Pepper	S3 (APT56, GCR50)	25.9	+7.27	+19.57	+1.53	−8.73	+13.25	+3.02	+11.19	611.7
Pepper	S3 (APT52, GCR65)	36.0	+8.16	+27.56	+0.04	−13.93	+17.99	+4.64	+15.19	870.2
Pepper	S3 (APT44, GCR65)	40.8	+7.43	+30.24	−1.36	−16.87	+19.69	+5.27	+15.87	1043.7
Pepper	S5 (APT44, GCR65, irrig.)	40.8	+12.25	+49.68	−1.36	−17.06	+21.56	+8.40	+9.02	1043.7
Pepper	S2 (c-Si, GCR40)	40.0	+6.95	+27.89	−0.78	−15.72	+17.90	+4.75	+12.11	1257.2
Tea	S3 (APT56, GCR50)	25.8	+0.01	+11.26	−3.47	−6.99	+4.73	+2.48	+8.29	652.5
Tea	S3 (APT44, GCR65)	40.7	−3.35	+15.98	−8.84	−13.53	+6.84	+3.86	+11.92	1113.3
Tea	S5 (APT44, GCR65, irrig.)	40.7	+2.47	+41.38	−8.84	−13.90	+8.36	+8.50	+0.74	1113.3
Tea	S2 (c-Si, GCR40)	40.0	−2.30	+16.56	−7.65	−12.63	+6.65	+3.68	+10.77	1374.4

3.3. Cooperative dispatch, dual-FU LCA, and break-even frontiers

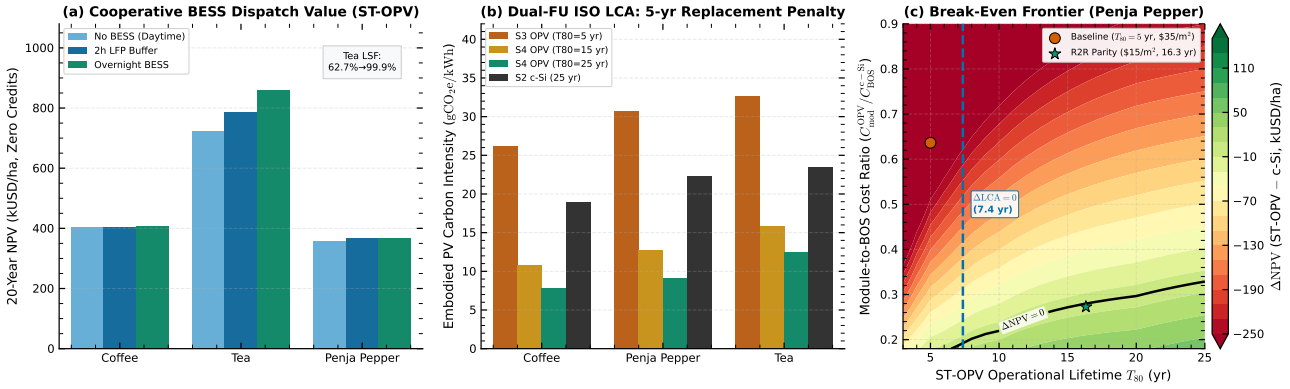


Figure 4: Techno-economic dispatch, environmental carbon intensity, and parity thresholds (0 USD tCO$_2$e^{-1}, 0 % subsidy baseline). (a) Effect of LFP storage sizing on 20 yr NPV for S_3(APT44, GCR65), highlighting Djuttitsa tea load solar fraction (LSF) climbing from 62.7 % to 99.9 %. (b) Embodied PV carbon intensity across 5 yr (S_3), 15 yr / 25 yr encapsulated ST-OPV (S_4), and 25 yr c-Si (S_2). (c) Two-dimensional NPV and LCA parity frontier over ST-OPV operational lifetime (T_{80}) and module-to-BOS cost ratio for Penja white pepper.

Integrating cooperative processing dispatch fundamentally differentiates Djuttitsa tea from daytime-dominated coffee and pepper cooperatives (Figure 4a, Table 4). Because tea withering consumes 45 % of its 307.8 MWh yr^{-1} factory electricity overnight, shifting from daytime-only operation without storage to overnight-withering LFP storage boosts load solar fraction from 62.68 % to 99.92 % and raises 20 yr NPV from 724 075 USD ha^{-1} (IRR = 38.65 %) to

860 825 USD ha^{-1} (IRR = 35.52 %, DPBP = 3.01 yr, LCOE = 0.0886 USD kWh^{-1}), delivering 393.2 tCO_2e ha^{-1} yr^{-1} in net areal carbon abatement after accounting for amortized embodied and on-farm diesel emissions (equivalent to 427.3 tCO_2e ha^{-1} yr^{-1} in gross displaced grid and diesel electricity). By contrast, Foumbot coffee and Njombe-Penja pepper cooperatives run primarily during daytime hours (LSF = 88.6 to 92.4 % without storage), gaining negligible incremental value from overnight BESS (NPV = 402.5 to 406.9 kUSD ha^{-1} coffee, DPBP = 3.95 yr with 2 h buffer; 358.0 to 368.4 kUSD ha^{-1} pepper, DPBP = 5.73 yr vs. 3.52 yr and 3.80 yr on elevated c-Si).

Under a dual-functional-unit ISO 14040/14044 assessment (T_{80} = 5 yr baseline; Figure 4b), frequent ST-OPV replacement drives embodied intensity (26.1 to 32.7 gCO_2e kWh^{-1}) above single-install 25 yr c-Si (18.9 to 23.5 gCO_2e kWh^{-1}). Extending encapsulated NFA lifetime breaks environmental parity cleanly at $T^*_{80,\mathrm{LCA}}$ = 7.42 yr (coffee), 7.40 yr (Penja pepper), and 8.12 yr (tea). Under S_4 encapsulation (T_{80} = 15 to 25 yr at baseline CAPEX), ST-OPV carbon intensity falls by 51 to 70 % to 10.8 to 15.9 gCO_2e kWh^{-1} at 15 yr and 7.8 to 12.5 gCO_2e kWh^{-1} at 25 yr (less than half c-Si intensity), LCOE declines by 23.1 to 33.8 % down to 0.0422, 0.0497 and 0.0646 USD kWh^{-1}, DPBP tightens to 3.76, 3.94 and 2.91 yr, and annualized net areal abatement rises to 334.5 tCO_2e ha^{-1} yr^{-1} (coffee), 296.7 tCO_2e ha^{-1} yr^{-1} (pepper), and 412.5 tCO_2e ha^{-1} yr^{-1} (tea). However, economic parity is far more stringent (Figure 4c): at current 35 USD m^{-2} module pricing, ST-OPV never reaches financial parity with c-Si within 30 yr ($T^*_{80,\mathrm{NPV}} > 30$ yr across all three crops) because elevated c-Si generates 21 to 24 % more electricity per hectare (1257 to 1479 MWh ha^{-1} yr^{-1} vs. 1044 to 1223 MWh ha^{-1} yr^{-1}). Only once roll-to-roll manufacturing compresses module CAPEX to 15 USD m^{-2} ($C^{\mathrm{OPV}}_{\mathrm{mod}}/C^{\text{c-Si}}_{\mathrm{BOS}} = 0.273$) does NPV parity emerge at $T^*_{80,\mathrm{NPV}}$ = 16.33 yr for Penja white pepper and 31.27 yr for coffee, whereas tea remains c-Si favored ($T^*_{80,\mathrm{NPV}} > 30$ yr) due to high captive electricity value per installed W_p.

Table 4: Techno-economic and life-cycle comparison (S_3(APT44, GCR65) vs. S_2(c-Si, GCR40)) and break-even thresholds at the strict zero-subsidy baseline (0 USD tCO_2e^{-1}, WACC = 10 %).

Crop	**System / Dispatch**	**LSF** (%)	**LER** (–)	**NPV** (kUSD ha^{-1})	**IRR** (%)	**LCOE** (USD kWh^{-1})	**LCA** (gCO_2e kWh^{-1})	**Off-Grid NPV** (kUSD ha^{-1})
Coffee	ST-OPV (no BESS)	92.42	1.401	402.5	26.46	0.0625	25.90	–
Coffee	ST-OPV (2 h BESS)	100.00	1.401	403.5	26.16	0.0637	26.16	−379.8
Coffee	c-Si (2 h BESS)	100.00	1.507	779.3	34.77	0.0380	18.92	–
Penja Pepper	ST-OPV (no BESS)	88.62	1.548	358.0	24.72	0.0733	30.36	–
Penja Pepper	ST-OPV (2 h BESS)	99.97	1.548	368.4	24.74	0.0750	30.73	−281.5
Penja Pepper	c-Si (2 h BESS)	99.99	1.640	709.1	32.55	0.0449	22.31	–
Highland Tea	ST-OPV (no BESS)	62.68	1.429	724.1	38.65	0.0687	28.46	–
Highland Tea	ST-OPV (2 h BESS)	74.30	1.429	785.7	39.02	0.0727	29.32	–
Highland Tea	ST-OPV (overnight BESS)	99.92	1.429	860.8	35.52	0.0886	32.66	+319.8
Highland Tea	c-Si (overnight BESS)	100.00	1.548	1272.6	41.66	0.0555	23.46	–

3.4. Sobol uncertainty quantification and policy stress tests

Across N = 10 000 Saltelli quasi-Monte Carlo parameter realizations spanning the full 12-parameter policy and techno-economic uncertainty box (carbon pricing $[0, 25]$ USD tCO_2e^{-1}; Figure 5a), ST-OPV exceeds elevated c-Si NPV in only 22.99 % of Penja white pepper draws (median −105.3 kUSD ha^{-1}, p05–p95 −394.4 to +116.9 kUSD ha^{-1}), 16.75 % of Arabica coffee draws (median −139.3 kUSD ha^{-1}, −437.6 to +90.2 kUSD ha^{-1}), and 11.00 % of tea draws (median −176.0 kUSD ha^{-1}, −490.0 to +58.0 kUSD ha^{-1}), meaning elevated c-Si dominates in 77.0 to 89.0 % of realizations. Under strict zero-carbon pricing (0 USD tCO_2e^{-1}), ST-OPV exceedance rises only slightly to 24.71 % for pepper (median −97.9 kUSD ha^{-1}), 18.51 % for coffee (median −131.0 kUSD ha^{-1}), and 12.34 % for tea (median −166.5 kUSD ha^{-1}), confirming that carbon credits slightly favor higher-output c-Si rather than rescuing ST-OPV ($S_{Ti} \leq 0.0011$).

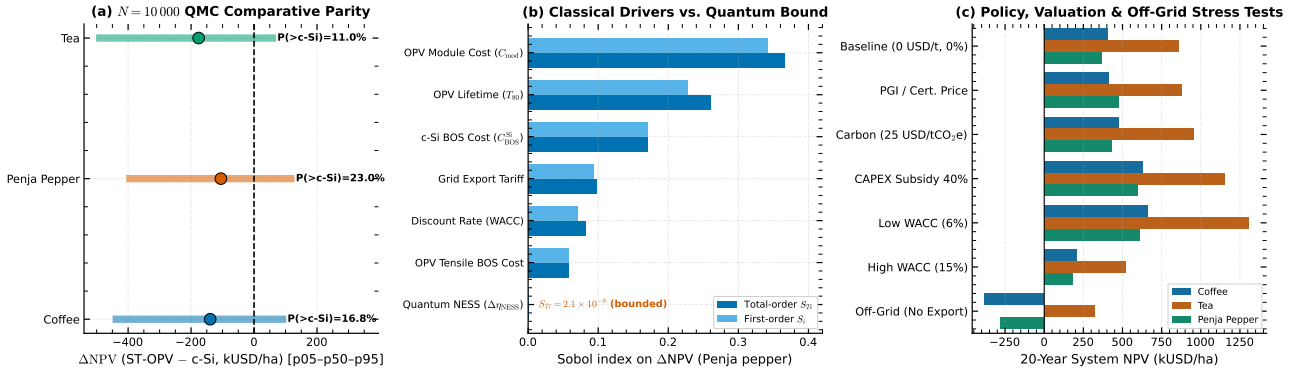


Figure 5: Probabilistic parity and global variance attribution ($N = 10\,000$ Saltelli QMC draws). (a) Comparative $\Delta\text{NPV} = \text{NPV}_{\text{OPV}} - \text{NPV}_{\text{Si}}$ [p05–p50–p95] distributions and exceedance probabilities. (b) Ranked first-order (S_i) and total-order (S_{Ti}) Sobol indices on Penja white pepper ΔNPV alongside the bounded quantum NESS index ($S_{Ti} = 2.10 \times 10^{-8}$; $\leq 2.58 \times 10^{-7}$ under the exact-HEOM envelope). (c) Policy, valuation, discount-rate, and off-grid stress tests across the three crops (S_3(APT44, GCR65)).

Contrast of Saltelli–Sobol total-order indices between standalone ST-OPV profitability (NPV_{OPV}) and comparative parity (ΔNPV) exposes a clear Dual Policy Bifurcation (Table 5, Figure 5b). Even though standalone ST-OPV installations remain consistently profitable across policy realizations (p05–p50–p95 spanning 145.0, 562.3 and 1116.5 kUSD ha^{-1} in coffee; 148.4, 537.8 and 1038.5 kUSD ha^{-1} in pepper; and 505.6, 1042.8 and 1780.2 kUSD ha^{-1} in tea), project bankability in daytime-dominated coffee and pepper is governed by grid export tariff ($S_{Ti} = 0.5176$ and 0.4263) and cost of capital WACC ($S_{Ti} = 0.2866$ and 0.3198), whereas energy-intensive tea processing reverses this hierarchy to WACC ($S_{Ti} = 0.4503$) and displaced diesel price ($S_{Ti} = 0.2817$) ahead of export tariff (0.1471). Once technology selection ΔNPV is evaluated at equal shade, shared macroeconomic parameters largely cancel, transferring over 74 % of cumulative variance to three hardware levers: OPV module CAPEX ($S_{Ti} = 0.323$ to 0.366, peaking at 0.3663 in pepper), ST-OPV T_{80} lifetime ($S_{Ti} = 0.247$ to 0.267), and c-Si elevated steel BOS CAPEX ($S_{Ti} = 0.150$ to 0.170).

Table 5: Dual Saltelli–Sobol decomposition ($N = 10\,000$ base samples, 140 000 evaluations per crop) contrasting macro-financial drivers of standalone bankability (NPV_{OPV}) against hardware engineering drivers of comparative parity ($\Delta\text{NPV} = \text{NPV}_{\text{OPV}} - \text{NPV}_{\text{Si}}$).

Crop	$\text{NPV}_{\text{OPV}}^{p50}$ [p05, p95] (kUSD ha^{-1})	**Top** $S_{Ti}(\text{NPV}_{\text{OPV}})$ **drivers** (parameter, index)	$P(\Delta > 0)$ (%)	ΔNPV_{p50} (kUSD ha^{-1})	**Top** $S_{Ti}(\Delta\text{NPV})$ **drivers** (parameter, index)	S_{Ti}^{Q} (NPV_{OPV} / ΔNPV) (–)
Arabica coffee	562.3 [145.0, 1116.5]	Tariff (0.518), WACC (0.287), $C_{\text{mod}}^{\text{OPV}}$ (0.104)	16.75	−139.3	$C_{\text{mod}}^{\text{OPV}}$ (0.343), T_{80} (0.247), $C_{\text{BOS}}^{\text{c-Si}}$ (0.159)	1.16×10^{-9} / 3.83×10^{-9}
Penja white pepper	537.8 [148.4, 1038.5]	Tariff (0.426), WACC (0.320), $C_{\text{mod}}^{\text{OPV}}$ (0.125)	22.99	−105.3	$C_{\text{mod}}^{\text{OPV}}$ (0.366), T_{80} (0.260), $C_{\text{BOS}}^{\text{c-Si}}$ (0.170)	7.15×10^{-9} / 2.10×10^{-8}
Highland tea	1042.8 [505.6, 1780.2]	WACC (0.450), Diesel (0.282), Tariff (0.147)	11.00	−176.0	$C_{\text{mod}}^{\text{OPV}}$ (0.323), T_{80} (0.267), $C_{\text{BOS}}^{\text{c-Si}}$ (0.150)	1.89×10^{-9} / 9.85×10^{-9}

Policy stress tests (Figure 5c) reinforce this dispatch-dependent bifurcation under grid isolation: removing grid export drops NPV to −379.8 kUSD ha^{-1} in coffee and −281.5 kUSD ha^{-1} in Penja pepper because captive spice and pulping demand absorbs < 10 % of array output, whereas Djuttitsa tea stays strongly profitable at +319 794 USD ha^{-1} (IRR = 20.06 %, DPBP = 6.61 yr). Certified PGI pricing (14 000 XAF kg^{-1}) adds $\sim$ 100 kUSD ha^{-1} to Penja white pepper NPV, while concessional capital (WACC = 6 %) matches a 40 % upfront CAPEX grant in raising system NPV across all basins.

4. Quantum-to-system boundary analysis: pulsed coherent dynamics vs. CW sunlight

Answering the second half of our central research question requires establishing precisely where coherent quantum biology matters and where it rigorously decouples. On ultrafast pulsed timescales representative of 2D electronic spectroscopy, non-Markovian bath memory,

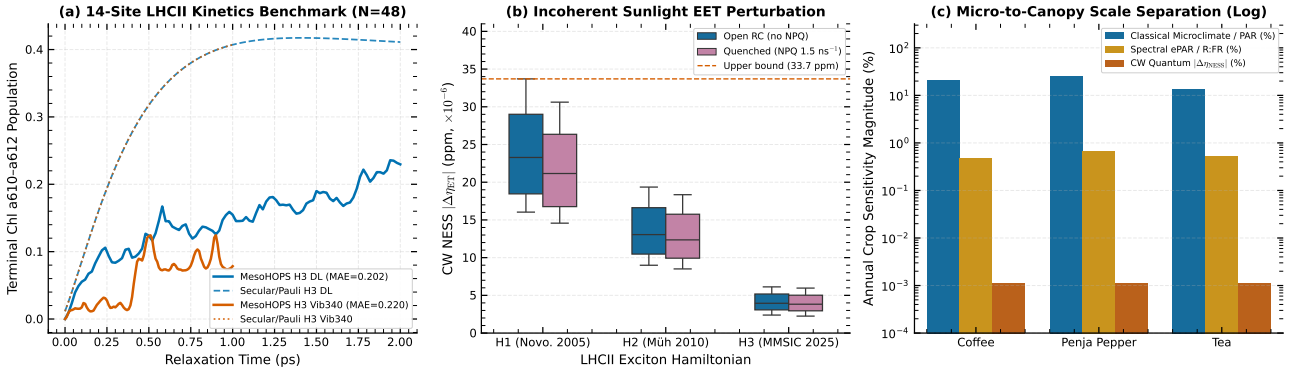


Figure 6: Microscopic open-quantum light-harvesting validation and canopy scale separation. (a) Full non-Markovian 14-site MesoHOPS 1.8.0 pulsed excitation kinetics ($N = 48$ disorder draws, $T = 295\,\mathrm{K}$) versus diagonal secular Pauli kinetics on H3 (Drude–Lorentz bath alone to 2.0 ps and augmented with the $340\,\mathrm{cm}^{-1}$ vibronic mode to 1.0 ps). (b) Continuous-wave (CW) solar NESS perturbation $|\Delta\eta_{\mathrm{ET}}|$ across all ST-OPV transmittance spectra (0° beam and diffuse) and three independent LHCII Hamiltonians (H1–H3), strictly bounded at 33.7 ppm (3.37×10^{-5}). (c) Log-scale comparison of annual canopy crop sensitivity magnitudes ($\sim 10\,\%$ microclimate vs. $\sim 10^{-3}\,\%$ maximum quantum EET shift).

environment-assisted quantum transport (ENAQT), and engineered laser excitation [67, 68, 69, 70, 71, 72, 36], our 14-site MesoHOPS 1.8.0 benchmark ($N = 48$ seeds; Figure 6a) confirms substantial transient deviations between non-Markovian coherent dynamics and secular Pauli transport: terminal Chl a_{610}–a_{612} site population exhibits MAE = 0.0198 (H2–DL), 0.2017 (H3–DL), and 0.2200 when the $340\,\mathrm{cm}^{-1}$ vibronic mode is included (H3–DL+Vib_{340}, reaching $p_{\mathrm{term}} = 0.0835$ vs. 0.4035 Pauli at 1.0 ps).

Under natural, temporally incoherent continuous-wave (CW) solar pumping filtered through PM6:Y6 stacks, however, inter-excitonic phase coherences decay orders of magnitude faster than solar photon arrival times [42, 37]. Solving exact Bloch–Redfield partial-secular NESS across all ST-OPV transmittance spectra and three independent LHCII Hamiltonians ($N = 200$ realizations; Figure 6b) yields a tight upper bound on coherent efficiency perturbation:

$$\max|\Delta\eta_{\mathrm{NESS}}| = 3.370 \times 10^{-5}\ \ (\mathrm{H1}), \qquad 1.936 \times 10^{-5}\ \ (\mathrm{H2}), \qquad 6.125 \times 10^{-6}\ \ (\mathrm{H3}), \tag{3}$$

under open reaction centers ($k_{\mathrm{RC}} = 0.005\,\mathrm{ps}^{-1}$), dropping further to 3.06×10^{-5} (H1) under NPQ quenching ($1.5\,\mathrm{ns}^{-1}$) and staying $\leq 1.181 \times 10^{-5}$ across $k_{\mathrm{RC}} \in [0.0033, 0.5]\,\mathrm{ps}^{-1}$ (validated against 4-site exact QuTiP-BoFiN HEOM [49] within 2.10×10^{-3}). Under our conservative one-to-one mapping onto crop yield and widened 30–60-fold uncertainty intervals (Section 2.1), the resulting total-order Sobol sensitivity of comparative parity ΔNPV to quantum NESS modulation reaches only $S_{Ti} = 2.10 \times 10^{-8}$ (3.83×10^{-9} coffee, 2.10×10^{-8} Penja pepper, 9.85×10^{-9} tea; Figure 5b, and $\leq 7.15 \times 10^{-9}$ on standalone $\mathrm{NPV}_{\mathrm{OPV}}$)—roughly seven orders of magnitude weaker than classical microclimatic and economic levers (Figure 6c). Even when widened further across the full exact-HEOM envelope ($\Delta\eta_{\mathrm{NESS}} \in [-0.21, +0.21]\,\%$), Sobol sensitivity remains $S_{Ti} \leq 2.58 \times 10^{-7}$ (2.60×10^{-7} at zero carbon), while exceedance probabilities $P(\mathrm{NPV}_{\mathrm{OPV}} > \mathrm{NPV}_{\mathrm{Si}})$ shift by ≤ 0.02 percentage points. This bounded scale-separation analysis cautions against attributing macroscopic agrivoltaic yield trends to steady-state photosynthetic quantum coherence.

5. Discussion

5.1. Generalizable deployment rules and tropical transferability

Synthesizing the coupled modeling components yields conditional biophysical screening rules for distinguishing potentially compatible from counterproductive tropical agrivoltaic configurations. Unlike arid Mediterranean or Sahelian installations where persistent clear skies prevent

wet-season PAR starvation [13, 6], humid–dry tropical highlands pair extreme dry-season radiative and evaporative stress ($\mathrm{PAR}_{95}/I_{\mathrm{sat}} = 1.91$ to 2.45, $\mathrm{VPD}_{95} = 2.83$ to 4.07 kPa) with persistent monsoon cloudiness (diffuse sky fractions of 0.42 to 0.50). Under fixed overhead PV canopies, therefore, net annual rainfed compatibility depends on whether dry-season photochemical and thermal relief (Figure 3b) outweighs overcast light limitation. Among the three modeled crops, PGI Penja white pepper has the most favorable rainfed yield response in the 2023 baseline; its modeled photoinhibition threshold ($I_{\mathrm{PI}} = 900\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$) maximizes dry-season photo-protective value (+30.24 % under S_3(APT44, GCR65)), and the modeled opportunity to replace annual dead or living tutors ($350\,\mathrm{USD\,ha^{-1}\,yr^{-1}}$) with an overhead agrivoltaic trellis accompanies annual productivity gains in this baseline (+7.27 to +8.16 %, LER = 1.48 to 1.55) and the highest baseline parity odds (23.0 %). Other trellis-climbing specialty perennials, including black pepper, vanilla, and passionfruit, are candidates for site-specific screening; their crop response and economic benefit cannot be inferred from the Penja white-pepper result without species- and site-specific validation. Conversely, fixed 40.6 % PV shade penalizes Foumbot Arabica coffee (−7.08 %, nearly three times steeper than agroforestry S_1 at −2.35 %) and highland tea (−3.35 %) because unshaded monsoon diffuse sky fractions already suppress ambient PAR. For the modeled Cameroon sites, supplemental dry-season deficit irrigation (S_5, restoring coffee to +0.22 % and tea to +2.47 %) is preferable to motorized monsoon retraction in the modeled comparison. Applying this rule to East African, Andean, or South Asian highland Arabica and tea estates requires local climate, crop, and energy-system validation. A quantified June–October full-opening counterfactual (S_6, Table S5) rescues yields by +1.22 to +7.38 pp (S_3) and +0.98 to +6.98 pp (S_2) but forfeits 38.0 to 40.1 % of annual electricity generation, eroding 20 yr NPV by 349 596 to 678 246 USD ha^{-1} for ST-OPV (425 667 to 784 209 USD ha^{-1} for c-Si) and leaving zero (0.00 USD m^{-2}) capital headroom for motorized actuation wherever grid-connected power monetization anchors bankability.

5.2. Materials engineering, cooperative dispatch, and policy governance

At the device-engineering scale, matched-shade counterfactual comparison overturns a central premise of wavelength-selective agrivoltaics: shifting PM6:Y6 bandgaps or APT changes annual crop yields by at most 1.05 percentage points relative to elevated c-Si ($\phi_{\mathrm{spec+Q}} = -0.21$ to -0.67 %) because PM6:Y6's 700 nm band edge trims transmitted ePAR/PAR slightly below unfiltered sunlight. Even though low NIR absorption halves daytime ST-OPV self-heating ($\Delta T_{\mathrm{PV}} = 3.56$ to 4.29 °C vs. 7.73 to 9.47 °C for c-Si) and protects module PCE via minimal thermal derating (−0.008 % °C^{-1}), total shade fraction governs leaf cooling (ΔT_{leaf} within 0.29 °C) and carbon assimilation. Accordingly, tropical ST-OPV materials research should pivot from narrow-bandgap optical tailoring toward moisture/UV barrier encapsulation and manufacturing cost reduction: crossing environmental parity requires $T^{*}_{80,\mathrm{LCA}} \geq 7.40$ to 8.12 yr (which under S_4 slashes carbon intensity below half c-Si, down to 7.8 to 12.5 $\mathrm{gCO_2e\,kWh^{-1}}$), whereas economic NPV parity demands roll-to-roll module pricing ($\leq 15\,\mathrm{USD\,m^{-2}}$) alongside $T^{*}_{80,\mathrm{NPV}} \geq 16.33$ yr in Penja pepper. Simultaneously, cooperative electricity demand topology dictates storage architecture: overnight tea withering (45 % nighttime load) turns LFP storage into a value-creating resilience asset (LSF = 99.92 %, DPBP = 3.01 yr, net areal abatement 393.2 $\mathrm{tCO_2e\,ha^{-1}\,yr^{-1}}$, and +319.8 kUSD ha^{-1} NPV even off-grid), whereas daytime coffee and pepper loads lose viability under grid isolation (−281.5 to −379.8 kUSD ha^{-1}). Finally, the Dual Sobol Policy Bifurcation (Table 5) provides concrete institutional guidance: rural minigrid feed-in tariffs ($S_{Ti} = 0.43$ to 0.52 on $\mathrm{NPV_{OPV}}$ for coffee/pepper) and concessional finance (WACC = 6 %, matching a 40 % CAPEX grant) govern standalone cooperative bankability, whereas materials R&D ($C^{\mathrm{OPV}}_{\mathrm{mod}}$, T_{80}) and steel BOS pricing govern technology competition (ΔNPV). Alongside our bounded open-quantum analysis ($S^{Q}_{Ti} \leq 2.60 \times 10^{-7}$), these variance maps retire both speculative steady-state quantum leverage and modest carbon pricing

($S_{Ti} \leq 0.0052$) as primary policy instruments.

5.3. Methodological boundaries and field validation roadmap

Several structural simplifications bound these model-based inferences and define priorities for paired field validation. First, root-zone FAO-56 available water capacity (160 mm) and leaf boundary-layer conductance (1.2 $\mathrm{mol\,m^{-2}\,s^{-1}}$) rely on literature-calibrated surrogates rather than site-specific soil-core retention curves or eddy-covariance flux towers, and were not among the 12 parameters varied in the WP5 Sobol QMC ensemble. A dedicated 3×3 coupled WP3 audit (Table S4) across AWC $\in [120, 220]$ mm and $g_b \in [0.6, 2.5]$ $\mathrm{mol\,m^{-2}\,s^{-1}}$ demonstrates that g_b aerodynamic decoupling governs absolute yield signs (ΔY_{S_3} spanning −13.40 to +6.42 % in coffee, −4.42 to +29.17 % in pepper, and −8.21 to +6.13 % in tea), whereas soil AWC shifts yields by only 0.47 to 2.63 pp at baseline g_b and the ST-OPV versus c-Si spread stays bounded within −1.14 to +2.41 pp across all 27 corners. Second, meteorological forcing spans one baseline year (2023 ERA5, an El Niño onset year) at gridded spatial resolution; synthetic meteorological stress perturbations around 2023 ERA5 (Table S4) confirm that warm–dry anomalies ($\Delta T_{\mathrm{air}} = +1.5\,^{\circ}\mathrm{C}$, −25 % precipitation) amplify shade value (S_3 ΔY rising to +1.28 % in coffee, +16.67 % in pepper, and +1.67 % in tea), whereas cool–cloudy conditions ($\Delta T_{\mathrm{air}} = -1.0\,^{\circ}\mathrm{C}$, −10 % GHI) deepen light limitation (−12.82 % coffee, +0.74 % pepper, −7.77 % tea). Multi-year field lysimetry, eddy-covariance turbulent flux measurements, and local contractor tenders across wind-exposed ridges remain essential to narrow structural and meteorological uncertainty. Third, canopy geometry is held static and annual net CO_2 assimilation maps linearly onto marketable dry yield without resolving multi-year carbohydrate storage, flowering phenology, or humidity-mediated fungal epidemics under wet-season shade—specifically coffee leaf rust (*Hemileia vastatrix*) and pepper *Phytophthora* foot rot—warranting paired microclimate-pathology trials under shaded versus open plots. Finally, the quantum scale-bridging test evaluates continuous-wave steady-state excitonic efficiency shifts mapped proportionally onto assimilation and therefore cannot exclude ultrafast transient signaling or downstream photochemical feedback outside the LHCII antenna complex.

6. Conclusions

We answered our central two-part question across PM6:Y6 transfer-matrix optics, open-quantum LHCII dynamics, hourly Cameroon microclimate physiology, cooperative processing dispatch, replacement-aware LCA, and Dual Saltelli–Sobol uncertainty propagation. First, under wet–dry seasonality and zero-subsidy baseline economics, dry-season cooling raises rainfed assimilation across all three crops (+8.8 to +30.2 %), whereas monsoon light deficits turn annual yields negative in Foumbot Arabica coffee (−7.08 %) and Djuttitsa highland tea (−3.35 %) under 40.6 % shade, leaving PGI Penja white pepper (+7.43 to +8.16 %, LER = 1.48 to 1.55) as the sole perennial sustaining net rainfed gains. Comparing ST-OPV with elevated c-Si at matched ∼ 40 % shade isolates crop differences within 1.05 percentage points ($\phi_{\mathrm{spec+Q}} = -0.21$ to -0.67 %), establishing that shade fraction and seasonal timing dominate spectral selectivity. Overnight tea withering BESS lifts load solar fraction to 99.92 % and delivers 393.2 $\mathrm{tCO_2e\,ha^{-1}\,yr^{-1}}$ in net abatement alongside profitable off-grid returns (+319.8 $\mathrm{kUSD\,ha^{-1}}$). While ST-OPV breaks life-cycle carbon parity at $T^{*}_{80,\mathrm{LCA}} = 7.40$ to 8.12 yr (cutting intensity to 7.8 to 12.5 $\mathrm{gCO_2e\,kWh^{-1}}$ and LCOE by 23.1 to 33.8 % at 25 yr), elevated c-Si dominates comparative NPV across 77.0 to 89.0 % of baseline QMC draws (75.3 to 87.7 % at fixed zero carbon), requiring roll-to-roll module pricing ($\leq 15\,\mathrm{USD\,m^{-2}}$) alongside $T_{80} \geq 16.33$ yr for parity in Penja pepper. Dual Sobol attribution confirms that export tariffs and WACC govern standalone cooperative profitability ($\mathrm{NPV_{OPV}}$), whereas module CAPEX, operational lifetime, and steel BOS govern comparative technology choice (ΔNPV). Second, filtered solar illumination constrains excitonic NESS efficiency shifts below 3.37×10^{-5} ($\leq 2.10 \times 10^{-3}$ under an exact-HEOM envelope), bounding

financial sensitivity at $S_{Ti} \leq 2.60 \times 10^{-7}$ and establishing that classical shade timing, battery dispatch, and encapsulation cost–durability targets—not steady-state quantum coherence—govern tropical agrivoltaic deployment.

CRediT authorship contribution statement

Steve Cabrel Teguia Kouam: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Theodore Goumai Vedekoi:** Formal analysis, Validation, Writing – review & editing. **Jean-Pierre Tchapet Njafa:** Methodology, Visualization, Writing – review & editing. **Jean-Pierre Nguenang:** Supervision, Validation, Writing – review & editing. **Serge Guy Nana Engo:** Supervision, Resources, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used AI-assisted software tools for manuscript proofreading, LaTeX formatting, and code-documentation checks. After using these tools, the authors reviewed, verified, and edited all scientific content, numerical tables, and conclusions and take full responsibility for the content of the publication.

Data availability

All ERA5 hourly meteorological series, optical transfer-matrix spectra, open-quantum Hamiltonian and steady-state efficiency datasets, and coupled scenario outputs are archived in the Supporting Material and project repository.

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